

STARTER TASK

Program 01

Given an array of integers $a[0], \dots, a[n-1]$, where $n > 1$, the maximum subarray is a continuous fragment $a[i], \dots, a[j]$ with the maximum non-negative sum of elements calculated according to the formula:

$$\text{sum}(i, j) = 3 * D + 2 * U \quad \text{for } 0 \leq i \leq j < n$$

where D is the sum of positive elements of the subarray $a[i], \dots, a[j]$, and U is the sum of negative elements of the subarray $a[i], \dots, a[j]$.

For example, for the sequence 2, -7, 4, -8, 5, 6, the maximum subarray is $a[4..5]$ and has the value $\text{sum}(4, 5) = 33$. If all numbers are negative, the subarray is empty and the sum is 0.

Write a program containing an efficient method with complexity $O(n)$, calculating the maximum subarray, i.e. the beginning, the end, and the maximum value of the subarray: $[i, j, \text{sum}(i, j)]$.

Notes

- Write the main algorithms in functions.
- Remember to add comments to the code.

Input

The first input line contains an integer z - the number of data sets. The descriptions of the data sets follow one after another.

One data set has the following form:

- The first line contains an integer n ($1 \leq n \leq 10^6$), denoting the size of the array.
- The next n lines contain n integers, which are the elements of the array.
- The absolute values of the array elements do not exceed 2000.

Output

For each data set, print one line in the form: beginning, end, and maximum value of the subarray.

Example

Input

```
1
6
2
-7
4
-8
5
6
```

Correct output

```
4 5 33
```

Submission rules

The .cpp file(s) should be named:

```
Surname_Name_Program_01.cpp
```

The source files, all other created files, and test text files should be placed in the folder `Surname_Name_Laboratory_1_starting`.

The file should be submitted to the folder: Programs - laboratory 1 - starting task - Tuesday 07:30.

The folder should be packed as a .rar or .zip archive and submitted to the course folder for Programming Methods - laboratory 1_starting on delta.pk.edu.pl.